

MLOps

The engineering discipline for reliably deploying, monitoring and maintaining models in production

GLOSSARY 04 - SHEET 5 / 8

PROFESSIONAL DEFINITION

MLOps is the set of engineering practices for reliably deploying, monitoring, retraining and retiring machine-learning models in production. It extends DevOps by adding the two dependencies that distinguish machine-learning systems from conventional software: the behaviour of the system depends on data as well as on code, and the system degrades over time as the world it was trained on changes, without any change to the code (Sculley et al., 2015; Kreuzberger, Kuhl and Hirschl, 2023).

WHO DEFINED IT ORIGINALLY

The problem was characterised by D. Sculley and colleagues at Google in the 2015 NeurIPS paper on hidden technical debt in machine-learning systems. The term itself emerged from practitioner communities from around 2018; Dominik Kreuzberger, Niklas Kuhl and Sebastian Hirschl (2023) provided the first systematic academic definition and architecture in IEEE Access.

WHY THIS DEFINITION WAS DEVELOPED

Sculley and colleagues demonstrated that the model code is a small fraction of a real machine-learning system, and that the surrounding infrastructure - data collection, feature extraction, serving, monitoring - accumulates a distinctive form of technical debt. A discipline was needed because software engineering practice assumed that a passing test today implies a passing test next quarter, which is untrue of models.

FIGURE 1 - THE CONTINUOUS MODEL LIFECYCLE

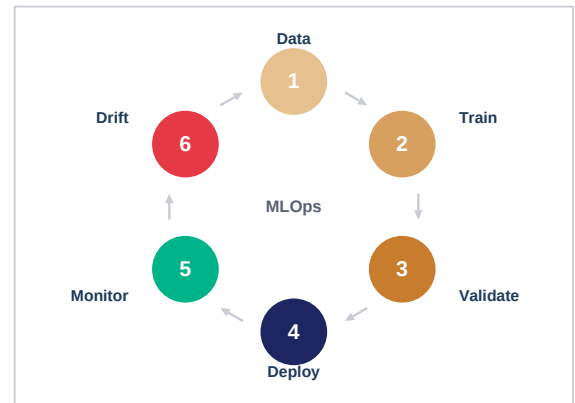


Figure 1. The loop does not close at deployment. Drift detection returns the system to data and retraining without a code change, which is the property that distinguishes MLOps from DevOps and the reason a released model still requires an owner (after Kreuzberger, Kuhl and Hirschl, 2023).

HOW IT IS USED

Version data and models alongside code, automate the retraining and deployment pipeline, monitor for data and concept drift in production, and define the retraining trigger and the rollback path before the first deployment.

WHEN IT IS USED

Before the first model reaches production rather than after the first incident, at every model release, when performance degrades without a code change, and at audit of a deployed high-risk system.

BY WHOM IT IS USED

Machine-learning and platform engineers, data science leads, model risk and validation functions, and the operations teams who carry the system after the project closes.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

DevOps unified development and operations for systems whose behaviour is fully determined by code, and its continuous-integration practice assumes deterministic tests. MLOps adds data and model versioning as first-class artefacts and introduces continuous training triggered by observed drift, which has no DevOps equivalent. Without it, a model that passed validation at release degrades silently, and the first indication is a business outcome rather than an alert.

RELATED DEFINITIONS IN THIS COURSE

Data Strategy - Unit 3

AI Operating Model - Unit 3

Proof of Concept and Pilot Purgatory - Unit 4

AI Governance - Unit 6

Explainability - Unit 6

SOURCES (HARVARD REFERENCING STYLE)

- Kreuzberger, D., Kuhl, N. and Hirschl, S. (2023) 'Machine learning operations (MLOps): overview, definition, and architecture', IEEE Access, 11, pp. 31866-31879.
- Sculley, D., Holt, G., Golovin, D., Davydov, E., Phillips, T., Ebner, D., Chaudhary, V., Young, M., Crespo, J.-F. and Dennison, D. (2015) 'Hidden technical debt in machine learning systems', Advances in Neural Information Processing Systems, 28, pp. 2503-2511.
- Treveil, M. and the Dataiku Team (2020) Introducing MLOps. Sebastopol, CA: O'Reilly Media.